A 208-state consistency machine

Comparison of the U209 proposals and a new conditional tape-context reduction

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Abstract

We construct an explicit binary, single-tape Turing machine with **208 working states**, including its blank initializer, by deleting one state from the supplied U209 candidates. The reduction changes only two retained transition rows. Its new ingredient is a conditional tape invariant: when an existing helper reads zero, its predecessor cell is one, whereas the deleted state has the two-cell predecessor word 10. Four exact local identities, of lengths $7, 1, 3, 1$, permit the helper to serve both roles. A five-state inductive invariant and 414 ordinary row identities prove halting equivalence on every common initial tape from state 0. For blank computations the equivalence is formalizable in PA; composition with the retained unary-theory calibration gives $\mathrm{PA} \vdash \neg\,\mathrm{Halt}(U_{208}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})$.

The six supplied submissions represent only two machines up to rooted labeled graph isomorphism. Both admit the new reduction, yielding two nonisomorphic 208-state variants. The primary candidate passes a fresh replay of all 5,084 symbolic primitive contracts, with all 416 rows covered, and retains the complete 48,140-step startup trajectory. The archive contains mathematical proofs and executable certificates; it does not contain a newly exported end-to-end proof-assistant certificate for the consistency biconditional.

1 Result and machine conventions

The primary artifact is `machine/u208.tm`. Its working states are $0, \ldots, 207$ and its separate halt label is 208. The tape alphabet is $\{0, 1\}$, the tape is indexed by $\mathbb{Z}$, and the initial configuration has state 0, head coordinate 0, and all cells zero. Every working-state/symbol pair has exactly one transition and every transition moves left or right. A transition into the halt label performs its write and move before stopping. The initializer is included in the 208-state count.

A table row `q b w D r` means: in state q, reading b, write w, move one cell in direction D, and enter state r. The header `#! start 0` is metadata, not a transition. The proofs use sparse *native* state names, in which 211 is a working state and 254 is halt. `machine/state_map.json` gives the complete bijection from retained native names to dense labels.

Property	Primary candidate
Working states / transition rows	208/416
Changed retained rows / deleted rows, relative to Joint U209	2/2
Primitive source occurrences / guarded ordinal control edges	3,909/5,083
Changed source addresses or primitive operands	0
Symbolic primitive contracts / covered literal rows	5,084/416
Maximum local time expansion per old transition	7
Exact unchanged blank startup length	48,140

The dense table has SHA-256

`a2fee9e41311c0d43aaac871ea411be24eada17e6b3633c55f40b437101144e1.`

The native table has SHA-256

`b348a466a2c316511e0037ea3824de29d4c6ad37338404fbc98a3968fdb25135.`

The deterministic builder, the independent local checker, and the full table in Appendix A identify the candidate independently of the prose. Its local reduction is proved without a consistency assumption. Actual nonhalting is equivalent to consistency of PA; neither unconditional consistency nor global state minimality is claimed.

The scope of novelty is explicit: this is a further reduction of the six supplied U209 submissions at the inspected repository snapshot. The sources did not contain the new $172 \mapsto 211$ fold. The inspection is not an exhaustive search of all unpublished machines or a priority claim against the entire literature.

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2 Comparison of the supplied proposals

2.1 Frozen scope and exact identities

The incoming directory was retrieved at ProveIt commit

`343da82d7384d10ec9fd1c17361fc923b2d1d117`.

It contained five executable U209 packages, a sixth U209 ZIP containing only a PDF, and the unary-theory archive [1, 2, 3, 4, 5, 6, 7]. The download of every item was checked against its Git blob identifier. `reference/source_inventory.json` retains the exact URLs, names, sizes, and blob identifiers; the comparison snapshots also record SHA-256.

Submission	Dense SHA-256 prefix	Class	Source depth
Plain U209	`65ccb606c8e9`	A	42
Compatibility	`bda91bd40880`	B	41
Compatible Folds	`5ded628d692f`	A	42
Composed	`3a69998d818b`	A	42
Joint, full package	`5ded628d692f`	A	42
Joint, PDF-only package	`65ccb606c8e9`	A	42

Here "class" means rooted deterministic transition-graph isomorphism preserving the initial state, halt, read symbols, write symbols, and directions. Compatible Folds and full Joint have byte-identical native and dense tables. Plain and Composed differ from them only by state renaming. The four class-A executable packages also have byte-identical source occurrence maps. The PDF-only Joint appendix was extracted independently: all 418 printed transition rows agree with Plain. It therefore identifies Plain's table, even though its title resembles the full Joint package.

There are consequently *two*, rather than five or six, distinct machines in the comparison. A different archive, table hash, or rotation recipe need not describe a different machine.

2.2 Independent isomorphism and source checks

A rooted read-labeled graph map is forced by following zero and one edges from state 0. An independent breadth-first procedure propagates that map, checking writes, directions, injectivity, and the distinguished halt label. Every working state is graph-reachable in every table, so a consistent map covers the whole machine. This is syntactic graph reachability; it does not assert that the actual blank computation visits every state.

In native labels, the nontrivial permutations are particularly small:

Plain to Joint or Compatible Folds:	(217 129 214)(215 204 216),
Plain to Composed:	(215 129 204 214 216).

All other labels are fixed, including initial state 0 and halt 254. The checker verifies the conjugacy of every row. For Plain versus Compatibility, the forced map has an explicit conflicting assignment; the witness is retained in `comparison/independent_comparison.json`. Its edge words are paths in a labeled graph, not claimed tape histories.

The two source layouts differ in exactly 720 addresses. An independent interval-based resolver verifies the same ordered 3,909 opcode/operand pairs and all 5,083 guarded ordinal control edges. Both have maximum consecutive-zero run six and 197 distinct source subtrees. Compatibility reduces maximum address depth from 42 to 41. A shallower source tree alone does not establish a better asymptotic runtime for the complete proof search.

2.3 What is shared, and what each package contributes

The U209 machines descend from one common U212 construction. Their joint layouts allow deletion of native states 180, 181, and 218, which had previously been removed separately in three U211 proposals. Class A uses representatives 163, 220, and 157. Compatibility instead uses 216 for state 180, with 220 and 157 serving the other two folds. All packages retain the arithmetic-entry repair at native row 171/0, redirecting the backward entry from 7 to 205. That repair is necessary because a source decoder also serves an arithmetic role.

The common reduction method exploits bits written immediately before a decoder entry. Opposite predecessor bits allow two control roles to share a state when an existing helper recovers the role from the tape. The source reassociations ensure that the required tags coexist without breaking the primitive control-flow graph or other interpreter entries. Different deleted labels therefore do not automatically give additive savings: the helper and address requirements can conflict.

Plain. This package gives a detailed ledger of 30 closed source rotations from U212, explicit local quotient cases, and careful treatment of the arithmetic entry. It distinguishes algebraic intermediate layouts from valid final interpreters: some intermediate addresses have seven-zero runs, while the final runtime proof uses the restored six-zero bound. It also supplies a useful obstruction: simply deleting the apparently dominated propositional row destroys the intended PA calibration. A Boolean interpretation using equivalence as implication validates the remaining calculus while keeping its target false. That is a reason to retain the row, not an unexplored one-state saving.

Compatibility. This is the genuinely different U209 family. Its eight-factor reassociation changes 720 addresses relative to class A and produces depth 41. Its separate construction, runtime, and theory audits make the changed source layout easy to inspect. It provides an important second search starting point, though its numerical calculus and symbolic inference engine are shared with the other packages.

Compatible Folds. Its staged 212-to-211-to-210-to-209 construction makes the coexistence of the old three folds explicit. It is a useful presentation of the same literal machine later delivered in the full Joint package. The extra exposition is valuable even though it is not an additional transition-graph class.

Composed. The grouped source proof and independent local-fold checker provide another view of class A. The machine itself is a permutation of Plain's decoder labels; its savings cannot be added to Plain's savings.

Joint, full package. This is the selected baseline for the new construction. Its direct 18-selector layout, compact complete ancestry, independent source-prefix auditor, and finite C++ executor make it convenient to extend and review. The selection is about its evidence organization, not a smaller state count than the other class-A presentations.

Joint, PDF-only package. The report is readable and its printed table is complete, but it does not add an executable verification package. Its table agrees with Plain, not byte-for-byte with full Joint. This distinction matters when binding a mathematical argument to a literal transition file.

2.4 Fresh comparison evidence and remaining common dependencies

All five executable packages passed their advertised quick replay commands in separate copies. Those commands check manifests, deterministic construction, local folds, source bindings, and

startup according to their documented scope. The six-table comparison was independently rerun under ordinary and optimized Python. The comparison phase did not rerun five copies of the full symbolic runtime derivation. Instead, the new primary U208 table received its own complete fresh derivation in Section 5.

The inherited runtime engines are essentially the same implementation under label and import changes. Multiple passing receipts should not be presented as independent logical foundations. The unary-theory interpretation, native numerical calculus, strong compiler, and protected startup argument are also shared. The new reduction preserves these components and has a short separate local proof.

2.5 Why the new fold was outside the old local search

The retained Joint search tested direct aliases through 15 steps and one-row folds using opposite unconditional predecessor tags through 17 steps. No result was found in those scopes. Native state 211 is entered from both directions and does not possess a uniform predecessor tag. Its useful property is conditional on reading zero; state 172 needs two predecessor cells. Extending the search to finite-history contexts exposes this interface. The successful proof needs only the five-state invariant and the four finite identities below, and does not depend on search bounds or random screening.

3 A conditional tape-context fold from 209 to 208 states

The reduction in this section acts directly on the supplied Joint U209 transition table. It preserves the tape alphabet, the blank initializer, and the entire numerical source program. Its additional information is a property of the recent computation: two states with different zero branches can share one physical state because their reachable tape contexts distinguish those branches. The required property of one state is conditional on its scanned symbol. The opposite-predecessor-bit interfaces considered in the supplied searches do not include this case.

All state labels in this section are *native* labels. In particular, 211 is a working state and 254 is the separate halt label. Dense renumbering is performed only after the reduction. Let U_{209} denote the Joint parent and U_{208} the reduced machine.

3.1 The complete modification

Delete state 172 and redirect its sole incoming transition to state 211. Change the zero transition of state 211 as follows:

Row	Parent action	Reduced action	Purpose
116/0	$0\,R\,172$	$0\,R\,211$	Redirect the deleted state
211/0	$0\,R\,157$	$0\,L\,217$	Inspect the preceding tape context

The removed rows are $172/0 : 0\,L\,67$ and $172/1 : 1\,R\,184$. Every other retained row is unchanged. Thus 209 working states and 418 transition rows become 208 working states and 416 rows. The initial state remains 0, and no halt transition is added or removed.

For the simulation, define the state map

$$f(q) = \begin{cases} 211, & q = 172, \\ q, & q \neq 172. \end{cases}$$

The map identifies two states, but changes neither the head position nor the tape. Its lack of injectivity is harmless: the proof follows a definite parent computation and establishes agreement at explicitly defined times.

3.2 Reachable contexts, proved by induction

A configuration is (q, h, T), where $h \in \mathbb{Z}$ and $T : \mathbb{Z} \to \{0, 1\}$. Write $t_j = T(h + j)$ for the tape viewed from the current head. The following five conditions are imposed only when the configuration has the indicated state:

$$I_{116}: \quad t_{-1} = 1, \qquad I_{172}: \quad t_{-2} = 1 \ \wedge \ t_{-1} = 0,$$
$$I_{217}: \quad t_{-1} = 1, \qquad I_{163}: \quad t_{-1} = 1 \ \vee \ t_{-2} = 1,$$
$$I_{211}: \quad t_0 = 1 \ \vee \ t_{-1} = 1.$$

Let $I(q, h, T)$ be the conjunction of the five corresponding implications from the state equality. In particular, I_{211} says that a scanned zero at state 211 forces the cell immediately to its left to be one. It does *not* assert that every entry to 211 has a left-hand one.

The entire incoming-edge census needed for these conditions is small:

Target	Every parent transition entering the target		
172	$116/0 : 0\,R\,172$		
116	$215/1 : 1\,R\,116$		
211	$163/0 : 0\,L\,211$,	$163/1 : 1\,R\,211$	
163	$132/1 : 1\,R\,163$,	$199/1 : 1\,R\,163$,	$217/0 : 0\,R\,163$
217	$216/1 : 1\,R\,217$		

These are inventories of all physical transition rows, including entries arising outside ordinary source decoding. The argument therefore does not depend on a sampled run or an assumption that additional entrances are unreachable.

Lemma 3.1 (Five-state invariant)**.** *Every configuration of U_{209} reached from state 0 satisfies I, on every initial binary tape.*

Proof. The initial configuration satisfies I because 0 is none of the five listed states. Assume I holds before a transition. Conditions whose target state is not reached impose no obligation afterward, so it suffices to inspect the displayed census.

An entry to 116 or 217 writes one and moves right, establishing its left-hand-one condition immediately. An entry to 172 comes from 116/0. It writes zero immediately left of the new head; the previous left-hand one supplied by I_{116} becomes the cell two positions left of that head. This proves I_{172}.

An entry to 163 from 132/1 or 199/1 writes one immediately left of the new head. Its remaining entrance, 217/0, writes a left-hand zero but carries the one supplied by I_{217} to the position two cells left. Each case establishes the disjunction I_{163}.

Finally, entry to 211 through 163/1 writes one immediately left of the new head. Entry through 163/0 moves left instead. The new current cell is the old t_{-1}, and the new left cell is the old t_{-2}. Their disjunction is exactly I_{163}, so I_{211} follows in this case as well. This proves preservation of all five conditions and completes the induction. □

The invariant explains the available distinction. State 172 necessarily arrives immediately to the right of the word 10. State 211, whenever it reads zero, instead has a one immediately to its left. Recovering this information needs at most three preceding parent transitions. Nevertheless, the inductive formulation is preferable for composition: it applies uniformly at every time, including the beginning of the run, without a separate assumption that a sufficiently long history already exists.

3.3 Four exact finite identities

At a simulation boundary the reduced machine has state $f(q)$, head h, and the same tape T as the parent. Only parent states 172 and 211 require special treatment. The four contracts below specify the complete effects of their replacement blocks. Head displacements are relative to h.

Parent state	Read	Additional tape condition	Steps	Final state, head
172	0	$t_{-2} = 1,\ t_{-1} = 0$	7	67, −1
172	1	None	1	184, +1
211	0	$t_{-1} = 1$	3	157, +1
211	1	None	1	184, +1

Every contract preserves the entire tape. The additional conditions for the zero cases follow from Lemma 3.1. The one cases need only the stated scanned bit, even though other invariant conditions may also hold.

Lemma 3.2 (Local replacement)**.** *Suppose a nonhalting parent configuration $C = (q, h, T)$ satisfies I, and its successor is $C' = (q', h', T')$. From $(f(q), h, T)$ the reduced machine reaches $(f(q'), h', T')$ in one, three, or seven transitions. All earlier configurations of this block are nonhalting.*

Proof. If $q \notin \{172, 211\}$, the parent and reduced rows agree after applying f to the target. This includes row 116/0, whose target is precisely the one redirected by f. There are 414 such ordinary rows.

For either exceptional state reading one, the reduced machine uses the unchanged row 211/1 : 1 *R* 184. Both parent states have that same row, so one transition suffices.

For parent state 211 reading zero, its invariant supplies $t_{-1} = 1$. The new row moves left into 217; row 217/1 : 1 *R* 215 moves back to the original zero; row 215/0 : 0 *R* 157 gives the endpoint of the parent's 211/0 transition. The three transitions read $0, 1, 0$ and write those same bits.

For parent state 172 reading zero, the invariant supplies $(t_{-2}, t_{-1}, t_0) = (1, 0, 0)$. The full reduced computation is:

Step	State before	Head before	Read/write	State after	Head after
1	211	0	0/0	217	−1
2	217	−1	0/0	163	0
3	163	0	0/0	211	−1
4	211	−1	0/0	217	−2
5	217	−2	1/1	215	−1
6	215	−1	0/0	157	0
7	157	0	0/0	67	−1

Its endpoint is exactly the parent transition 172/0 : 0 *L* 67. Every read is of one of the three specified cells and every write preserves that cell. The three-step case has the same property for its two cells. No unspecified exterior cell is inspected, and no exceptional block visits halt label 254. This proves all cases. □

The seven-step block visits the merged state 211 twice. There is no circular proof here: all seven transitions have been evaluated explicitly. In particular, the argument does not assume that arbitrary internal configurations of the reduced computation satisfy the parent invariant. The invariant is required only at parent simulation boundaries.

3.4 Global equivalence and an exact clock

For a parent computation, let $N_{172,0}(n)$ count occurrences of state 172 reading zero among its first n transitions, and define $N_{211,0}(n)$ similarly. Counts are taken along the parent run, not the internal states of the reduced blocks. Define

$$\tau(n) = n + 6N_{172,0}(n) + 2N_{211,0}(n). \tag{1}$$

The excesses six and two are the extra transitions used by the seven-step and three-step replacements.

Theorem 3.3 (Uniform halting equivalence)**.** *For every binary tape T and initial head position h, U_{209} and U_{208}, both started in state 0 on (h, T), either both halt or both run forever. At every parent time n up to and including its first halt, the reduced configuration at time $\tau(n)$ has the same head and tape, and its state is the image under f of the parent state. Furthermore,*

$$n \le \tau(n) \le 7n.$$

If the parent first halts at time H, the reduced machine first halts at exactly $\tau(H)$.

Proof. The configurations agree initially because $f(0) = 0$. Induct on the number of parent transitions. Lemma 3.1 supplies the required invariant at each parent configuration, and Lemma 3.2 extends the agreement by the appropriate positive block. The block lengths give (1). Each length is between one and seven, yielding the bounds. No block contains an earlier halt, so a parent first halt produces a reduced first halt exactly at the corresponding boundary.

If the parent never halts, the same induction supplies nonhalting reduced blocks at all boundaries. Their strictly positive lengths make $\tau(n)$ unbounded, and together the blocks cover every reduced time. The reduced machine therefore cannot halt between boundaries or after all of them. This also proves the converse implication of the halting equivalence. Determinism ensures that these blocks are prefixes of the actual reduced run, rather than merely possible executions. □

Thus the result concerns the whole computation, not only a finite prefix. It also supplies quantitative finite statements: a parent halt within n steps implies a reduced halt within $7n$ steps; a reduced halt within n steps implies a parent halt within n steps. No consistency assumption appears anywhere in the reduction.

3.5 Arithmetic scope and the other U209 family

The arbitrary-tape theorem above is a mathematical statement about tape functions. For the required blank input, configurations and finite runs have the usual arithmetic encodings. The five invariant checks, the four finite identities, and the induction on coded computation length are formalizable in PA. The explicit bound $7n$ removes any need for an unbounded search to locate a corresponding finite prefix. Consequently, this construction supplies the PA-formalizable equivalence

$$\mathrm{PA} \vdash \big(\mathrm{Halt}(U_{209}, \mathrm{blank}) \leftrightarrow \mathrm{Halt}(U_{208}, \mathrm{blank})\big).$$

Composing it with a justified parent consistency calibration transfers that calibration to U_{208}. The reduction neither establishes Con(PA) nor replaces the separate proof-search and theory interpretation obligations. The delivered mathematical proof and finite certificates do not constitute a newly exported proof-assistant certificate of the complete consistency endpoint.

The distinct Compatibility U209 family admits the same reduction: delete 172 into 211 and replace row 211/0 by $0\,L\,214$. Its local roles correspond as follows:

Role	Joint parent	Compatibility parent
Left-context selector	217	214
State feeding the merged-state entries	163	216
Right branch of the selector	215	163
Predecessor writing one before the selector	216	217

States 116, 172, 211, 157, 67, and 184 retain their roles. The corresponding complete censuses are: 172 from 116/0; 116 from 163/1; 211 from 216/0 and 216/1; 216 from 132/1, 199/1, and 214/0; and 214 from 217/1. Their directions and written bits are exactly those used in the invariant proof. The seven-step state sequence becomes

$$211 \to 214 \to 216 \to 211 \to 214 \to 163 \to 157 \to 67.$$

The same tape conditions, head positions, block lengths, and clock formula therefore apply. This produces a second 208-state table from the other supplied machine family, with no additional source transformation.

4 The retained consistency calibration

The new reduction preserves the entire source proof search. Its logical endpoint therefore comes from composing the preceding literal-machine equivalence with the retained calibration of the selected Joint U209 parent [6]. This section identifies the exact mathematical dependencies and the scope of their verification. The fixed K215 theorem about ZFC [8] is background and a source of reusable interfaces; it is not applied to the changed transition table by renaming its conclusion.

4.1 Unary power covering

The proposed theory U uses equality-free classical first-order logic over a nonempty domain, with one binary relation E, written as membership. Put

$$\mathrm{Sub}(x, a) := \forall t\,(E(t, x) \to E(t, a)).$$

For each formula φ whose free variables are contained in $\{x\}$, its sole nonlogical schema is

$$\forall a\,\exists b\,[\forall x\,(E(x, b) \to \varphi(x)) \;\wedge\; \forall x\,(\mathrm{Sub}(x, a) \to (\varphi(x) \to E(x, b)))]\,. \tag{2}$$

All bound variables are freshened before insertion. In particular, the source a cannot occur freely in φ. Arbitrarily many bound variables and unrestricted quantifier complexity are permitted. The cover contains all selected represented subsets of a, and may contain further selected objects. It need not be an exact filter of the subsets of a. The tempting exactification $\varphi(x) \mapsto \mathrm{Sub}(x, a) \wedge \varphi(x)$ is illegal in this unary schema.

The supplied report [1] proves interpretations in both directions between U and full PA, with uniform finite-proof transformations. The reviewed lower argument addresses the parameter restriction through a specific noncircular sequence of constructions, rather than assuming unrestricted separation.

Define coextensionality $u \simeq v$ by equality of member profiles. An object is saturated when its membership predicate respects this relation. Numerical representatives are saturated, transitive objects whose members are saturated and linearly ordered by membership modulo coextensionality. A further endpoint condition says that each nonempty intersection of an existing object's member profile with the number or one of its numerical initial segments has least and greatest member classes. All these conditions are fixed first-order formulas. They assert neither external finiteness nor external well-foundedness.

False filtering supplies an empty object. The local numerical facts give heredity, a greatest predecessor for each nonempty number, and uniqueness of existing successors modulo coextensionality. They do not initially assert that every number has a successor. A unary filter of failures then proves induction: if failures occur below a counterexample n, choose their least class and use its predecessor; if there are no failures below n, the predecessor of n supplies the contradiction directly. Neither case needs an already existing successor of n.

For an induction predicate $\theta(v, \bar{p})$, write $A_\theta(\bar{p})$ for its base and step premises. Parameter internalization uses the legal unary formula

$$Q(v) := \forall \bar{p}\, (A_\theta(\bar{p}) \to \theta(v, \bar{p})).$$

This yields induction with arbitrary ambient parameters, even when the predicate is not invariant under coextensionality. Induction next gives comparison of numerical classes and least numerical witnesses. Fixed finite tuples can be minimized lexicographically without assuming objects that already encode those tuples.

The resulting bounded comprehension theorem has explicit invariance and support hypotheses. Suppose $\chi(z, \bar{n})$ is invariant in its numerical parameters and every selected z is a subset of some supporting object. If exact comprehension fails, let $\bar{r}$ be the least failing numerical tuple. The unary matrix

$$\exists \bar{n}\, (\mathrm{LeastFailure}(\bar{n}) \wedge \chi(z, \bar{n}))$$

is extensionally $\chi(z, \bar{r})$. Filtering at a separately obtained support source gives the forbidden exact object. The parameters are bound inside the matrix. This is a finite classical proof construction, not an algorithm that evaluates a least failure or a hidden truth predicate.

This comprehension supplies total successors, powers of numerical sources, and saturated singleton, pair, and ordered-pair codes. Pair codes with coordinates in a number n are subsets of an exact power of n, providing supports for numerical graph comprehension. Deterministic histories are extended using a canonical formula that existentially binds the old history; an arbitrary graph is never inserted as a forbidden free matrix parameter. History uniqueness precedes extension, and a separate numerical bound controls values without assuming monotonicity. Successor iteration gives addition, iteration of addition gives multiplication, and the already derived induction gives every translated PA induction instance.

4.2 The upper interpretation and internal proof transport

For the reverse direction, interpret $E(x, a)$ as "bit x of a is one." PA proves that the translated subset condition implies $x \leq a$. For each fixed arithmetic formula χ, PA induction constructs its binary characteristic code on the interval through a. Applying this to $\mathrm{Sub}(x, a) \wedge \varphi(x)$ gives an exact filter and hence the weak cover (2). The argument also permits arbitrary extra matrix parameters. Because it uses PA's induction scheme, it applies in nonstandard PA models and requires no uniform satisfaction predicate. PA also proves global irreflexivity of this membership relation.

Once the lower interpretation is fixed, its finite non-induction obligations have fixed derivations; an arbitrary induction instance is inserted into the preceding uniform template. Fresh-name selection, substitution, operation-graph translation, relativization, and finite proof concatenation are primitive-recursive syntax operations. The upper translation similarly inserts each fixed translated formula into the finite-comprehension proof. Induction on formulas and coded derivations verifies the resulting refutation transformations in PA. Thus the supplied claim is the internal syntactic equivalence

$$\mathrm{PA} \vdash \mathrm{Con}(\mathsf{U}) \leftrightarrow \mathrm{Con}(\mathrm{PA}), \tag{3}$$

not merely standard-model equiconsistency. The package gives the ordinary mathematical uniformity argument; it does not provide a complete executable arbitrary-proof translator or a new proof-assistant certificate for this assertion.

4.3 What the native machine actually enumerates

The selected machine preserves the native Cantor-coded calculus. It does not newly compile the positive dyadic emitter also proposed in the unary report. Its ordinary machine-facing theory is $\mathsf{W} + \mathsf{IR}$: W allows arbitrary matrix parameters in the weak-cover schema, and IR is $\forall x\, \neg E(x, x)$. Its consistency calibration is the cycle of PA-verifiable reductions

$$\mathrm{Con}(\mathrm{PA}) \implies \mathrm{Con}(\mathsf{W} + \mathsf{IR}) \implies \mathrm{Con}(\mathsf{U}) \implies \mathrm{Con}(\mathrm{PA}).$$

The first arrow uses the bit interpretation, the second the inclusion of the unary instances, and the third the lower interpretation just reviewed.

The numerical calculus has truth alias 0, target 1, seven logical rows, a dominated propositional row, modus ponens, generalization, and a gated cover/vacuity bundle. In the retained role assignment, matrix code and output-variable label are a, the source label is b, and the tested label is c; the gate requires $c \neq a, b$. Every effective variable label in a positive matrix code is smaller than that code, while code zero is a closed truth alias. The output label is therefore fresh.

When $a \neq b$, the generated guard is the subset condition. When $a = b$, the guard's local binder captures the source occurrence and gives emptiness under irreflexivity. A cover at an empty source supplies this exceptional instance, including the small codes zero and one. The gate also preserves atomic membership vacuity. For completeness, a desired matrix is padded to an equivalent code $n > 1$ and emitted using $(a, b, c) = (n, 0, 1)$; the logical proof templates remove the padding and restore names. Auxiliary equality is handled through two-sided membership indiscernibility, not by silently assuming unrestricted primitive-equality vacuity. These are the retained U214/U213 calibration arguments in the parent's reference chain [6].

4.4 Proof search and the protected startup

The source contains 3,909 primitive occurrences. Its axiom command keeps the theorem-valued output and ten-entry stack interface needed by the three-command proof compiler. Strong completeness works above an arbitrary underlying numerical stack after counter and cache reset. Successful candidate codes are cofinal, and each residual continuation decreases; retaining a finite candidate cutoff therefore does not lose completeness. The native alias $P(0, 0) = 0$ is explicitly handled, including the special zero case of the counter-setting macro.

Startup needs a separate soundness argument. The shortened initializer reaches an interior axiom-command suffix, rather than a complete invocation. Its source bridge starts at ordinal 106 with logical store $(1, 0, 4, 0, 0, 1, 0, 0, 1)$ and reaches ordinal 136 after 141 source primitives. The raw cache value 1 is used as an arithmetic parameter, and the stack tail 4 is opaque; neither is assumed theorem-valued.

The suffix builds nine explicit theorem heads above that tail and reaches scheduler candidate 1 with counter and cache zero. A ghost explicit stack length is retained because the pairing alias prevents recovery of list length from its integer code. The all-natural scheduler-credit proof assigns conservative weights $-1, 0, 10$ to transfer, modus ponens, and axiom commands. Its maximal prefix debt from candidate 1 is two, leaving a permanent margin of seven heads. Hence no pop exposes the opaque tail. This unbounded argument supplies startup soundness; a finite blank trace alone would not do so.

4.5 Composed endpoint and proof status

Corollary 4.1. *For the exact delivered 208-state table, with its own initialization and blank input tape,*

$$\mathrm{PA} \vdash \neg\, \mathrm{Halt}(U_{208}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{U}) \leftrightarrow \mathrm{Con}(\mathrm{PA}).$$

Proof. Compose the new literal halting equivalence of Theorem 3.3 with the selected parent's retained numerical proof-search, compiler, startup, and theory calibration, then apply (3). □

No consistency assumption is needed for the biconditional. Asserting actual nonhalting additionally uses Con(PA). The result is an ordinary PA-formalizable mathematical proof, supported by exact finite checks and universal symbolic induction certificates. Those certificates are stronger than bounded register tests, but their checker is not a proof-assistant kernel. The separate formal K215/Con(ZFC) and Theory K developments [8, 9] establish different endpoints. This release does not claim a newly compiled end-to-end Lean, Rocq, or MM0 theorem for U208.

5 Verification of the delivered literal machine

5.1 Small independent certificate for the new theorem

The independent `code/check_local.py` imports neither the builder nor proposal code. It reads the pinned parent, independently constructs the prescribed child, and verifies its identity with both delivered native and dense tables. It uses explicit exceptions throughout, so its checks remain active under `python3 -O`.

The five-state invariant is checked as a finite Boolean implication for every possible local tape assignment needed by an incoming transition. The window $[-3, 1]$ suffices: a transition moves by one, and the farthest invariant clause looks two cells to the left. Eight transition rows enter states having nontrivial clauses. Their 256 Boolean valuations include 104 satisfying antecedents and 152 inapplicable cases. The other 410 old rows have a trivial destination clause. The initial clause is vacuous at state 0. This finite check, together with ordinary induction on computation length, proves the invariant for arbitrary tapes.

The checker separately verifies all 414 ordinary quotient-row equations and the four exceptional paths. An attempt to read an unspecified local cell is rejected; the outside tape is genuinely arbitrary. Dense renumbering supplies 416 further row conjugacy checks. Every one of the 208 working states is graph-reachable from the initial state.

Twelve deliberately damaged controls are rejected. They include an unredirected incoming edge, a wrong helper, a reversed selector direction, a changed write bit, corrupted helper branches, an incorrect endpoint direction, and removal of each needed zero-case context. Normal and optimized independent-checker receipts are byte-identical. A separate construction checker enumerates all three-step local histories, adding all shorter prefixes from the initial state; it independently recovers the two guards. The theorem is established by the explicit invariant and path proof, not by a sampled blank computation.

5.2 Fresh universal primitive replay

The source and its addresses are unchanged. Nevertheless, we reran the inherited whole-tape symbolic induction engine on the final 208-state literal table. This checks the primary candidate directly, in addition to the local reduction theorem. All 5,084 primitive contracts closed, using 1,251,865 symbolic inference applications; every one of the 416 literal rows was covered. The smallest constant duration was five, and all duration coefficients were nonnegative.

The symbolic frame has nine independently arbitrary natural register values, an arbitrary natural header gap, and an arbitrary nonempty finite spare-register list. Thirteen affine slots represent the constant, gap, nine primary values, first spare, and remaining list width. Four one-block traversal identities are rederived from the actual literal table; list induction and the unary self-loop rule extend them to arbitrary block lengths and finite lists. No finite register cutoff is used in these contracts. The canonical header distance is 56.

Each source increment, positive or zero decrement, branch, continuation, and halt is checked against its exact whole-tape effect. For nonhalting primitives the endpoint includes the next encoded source address, head displacement, all primary values, the spare list, and the blank exteriors. The 4,000-inference search cap per primitive is a limit on certificate discovery: every

reported case successfully closes with an explicit finite derivation. It is not a bound on register values or machine runtime.

These contracts are executable mathematical induction evidence. Their soundness depends on the stated unary/list rules and their implementation; they are not proof-assistant kernel objects. The new local proof requires only the much smaller invariant and finite path checker and does not depend on the universal engine to justify the fold itself.

5.3 Initializer and independent finite regression

The parent and new machine execute exactly the same first 48,140 transitions from blank tape. The full trace SHA-256 is

`41e9a165abae389c3b182ebbf8d8fd1c260d3d36d4f7ebb6d60e235bac40f19a.`

Both arrive in native state 84 with head -2, source ordinal 136, address `001100110101010`, and 120 one-cells. The logical register tuple is $[1, 0, 13, 0, 0, 1, 0, 0, 1]$ and the 43 spare registers all equal one. The checker verifies the complete finite set of one positions and both zero exteriors. No changed or deleted row is visited during startup. This binds initialization precisely; its soundness significance is the protected suffix argument in Section 4.

An independent C++ literal executor passed 15,252 primitive cases, covering three finite assignments for every guarded source branch and executing 6,778,830 primitive-level literal steps. It also ran a 2,000,000-step blank prefix without halting. The latter observations are regression evidence only; no finite prefix is offered as a proof of nonhalting or consistency.

Check	Fresh result and scope
Six-presentation comparison	Two rooted isomorphism classes; identical ordinal source graph
New inductive invariant	104 applicable local Boolean implications; arbitrary tapes by induction
New local simulation	414 ordinary rows and four exhaustive macros, maximum seven steps
Dense table and robustness	208 states, 416 rows, 12 damaged controls rejected; normal/$-O$ agreement
Universal runtime replay	5,084 contracts, all 416 rows, arbitrary natural values and finite lists
Exact startup	All 48,140 transitions and complete endpoint tape unchanged
Independent finite executor	15,252 primitive cases; supplementary finite evidence
End-to-end kernel certificate	Not produced for U208; inherited ordinary calibration retained

5.4 The alternative 208-state family

The Compatibility-derived table in `alternative/u208.tm` uses the same deletion with helper 214 and the corresponding old-role labels. It has 208 working states and 416 rows. Its dense SHA-256 is

`9865dcd7bd44deece098c16dcec7b90a96cf111592cbd37e3531b50f7b09276c.`

The alternate construction and its history, incoming-census, and local macro checks pass. We do not claim a separate full universal runtime rerun for this alternate table; its correctness transfers through its own exact local theorem from the supplied Compatibility parent.

The two new variants remain nonisomorphic. Let

$$p = \texttt{000000100100000001001101111},$$
$$q = \texttt{0000001001000000010011011001111111110}.$$

In the primary table both read-edge graph paths from state 0 end at native 98. In the alternate table they end at native 98 and native 215, respectively. A rooted read-edge-preserving bijection would have to preserve equality of these endpoints, which is impossible. These graph words need not be realizable as one physical tape history.

6 Reproduction, artifacts, and further work

The release is self-contained for its executable checks. Python 3.10 or later and the standard library suffice for the main replay. No network access, solver, optimizer, Lean, or Rocq installation is needed. From the unpacked top-level directory run

```
python3 -B reproduce.py
```

The default command verifies the release hashes, binds the selected parent and unchanged inference modules to the original Joint archive, runs that parent's quick construction/source/startup replay in a temporary directory, reconstructs both 208-state variants, checks the new local theorem normally and under optimized Python, checks the alternative's local histories, repeats the six-proposal comparison, verifies exact startup, and rederives all 5,084 primary U208 contracts. Generated temporary files do not alter the sealed release. To record a receipt, pass `-report` with a path outside the release directory.

```
python3 -B reproduce.py --quick
```

This command explicitly skips the fresh universal runtime derivation. It remains useful for the new small proof and exact construction, but must not be reported as a fresh full symbolic replay. The first complete symbolic run in this preparation environment took about 139 seconds; wall time is environment-dependent.

The optional independent finite executor requires a C++17 compiler:

```
g++ -std=c++17 -O3 code/check_finite_runtime.cpp -o /tmp/check_u208
/tmp/check_u208 machine/u208.native.tm machine/sites.tsv
```

Rebuild the PDF in a working copy, so generated TeX auxiliary files do not change the sealed inventory:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

PDF bytes may vary with the TeX version or metadata. Literal machine and source bytes are deterministic.

Path	Contents
`report.tex`, `report.pdf`, `paper/`	Editable report, proofs, comparison, full transition table
`machine/`	Primary native/dense U208 tables, state map, unchanged source
`alternative/`	Nonisomorphic Compatibility-derived U208 and its parent/source
`machine_parent/`	Exact selected U209 table and runtime label correspondence
`code/`, `reproduce.py`	Deterministic builder, independent invariant checker, runtime and startup replay
`comparison/`	All six table presentations, source snapshots, isomorphisms and fresh quick logs
`evidence/`	New local, startup, universal, finite, and release replay receipts
`reference/`	Full selected Joint archive with nested U212/U213/U214/U218/U222 lineage and unary theory; source provenance and reviews
`research/`	Discovery programs and precisely scoped search receipts
`licenses/`, `NOTICE.txt`	Upstream terms and attribution
`SHA256SUMS`	Hashes of all other sealed release files

6.1 What the new result suggests

The effective improvement is a stronger invariant language for state reuse. An unconditional incoming-bit test misses states entered from different directions. Here a two-cell old-mode tag and a conditional old-helper assertion suffice, even though the replacement path revisits the same physical helper. Complete backward history gives discovery candidates; small inductive assertions give a clearer proof.

A brief follow-up search on U208 found no further direct alias within seven-step history contexts and 27 replacement steps, nor a further shared-row fold across 348 ordered interfaces within the tested seven-history/25-step limits. These are restricted negative searches, not a lower bound at 208. Richer invariant combinations, several jointly changed rows, or high-level arithmetic changes remain possible. The report does not claim that any of those approaches already gives 207.

The immediate formalization target is compact: encode the five-state invariant and four finite trace identities in the preferred arithmetic or proof-assistant framework, then export the generic positive-block simulation theorem for the exact table. Completing an end-to-end U208 certificate also requires importing or formalizing the retained numerical-calculus and unary-theory proof transformations. That larger task should be distinguished from the new finite reduction already proved here.

References

[1] ProveIt incoming archive, *Unary power-cover theory U*, 29 October 2026. Pinned source archive. File `pa_unary_power_cover.zip`.

[2] ProveIt incoming archive, *U209 ConPA, Plain*, 29 October 2026. Pinned source archive. File `U209_ConPA_2026-10-29.zip`.

[3] ProveIt incoming archive, *U209 ConPA Compatibility*, 29 October 2026. Pinned source archive. File `U209_ConPA_Compatibility_2026-10-29.zip`.

[4] ProveIt incoming archive, *U209 ConPA Compatible Folds*, 29 October 2026. Pinned source archive. File `U209_ConPA_Compatible_Folds_2026-10-29.zip`.

[5] ProveIt incoming archive, *U209 ConPA Composed*, 29 October 2026. Pinned source archive. File `U209_ConPA_Composed_2026-10-29.zip`.

[6] ProveIt incoming archive, *U209 ConPA Joint, full package*, 29 October 2026. Pinned source archive. File `U209_ConPA_Joint_2026-10-29.zip`.

[7] ProveIt incoming archive, *U209 ConPA Joint, PDF-only presentation*, 29 October 2026. Pinned directory, file `U209_ConPA_Joint_2026-10-29(1).zip`.

[8] Elliot Glazer, *K215 halts iff ZFC is inconsistent: Metamath Zero, PA*. `https://github.com/elliotglazer/k215-pa`. Repository documentation inspected 29 October 2026; the large MM0 proof was not recompiled for this report.

[9] Elliot Glazer, *Theory K and the K230 Rocq development*. `https://github.com/elliotglazer/k230-rocq`. In particular, the Theory K section of the README.

[10] The `jeff87654/busy-beaver` reference repository, *ZF/m254 reference development*. `https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254`. See the original attribution notices preserved in this release.

A Complete primary transition table

Working states are $0, \ldots, 207$, halt is 208, and the initial state is 0. The head starts at coordinate 0 on the all-zero tape. Each entry lists the written symbol, move, and next state. Both table panels use these same conventions.

q	Read 0	Read 1	q	Read 0	Read 1
0	1 R 92	1 R 0	104	0 R 21	1 R 120
1	0 L 1	1 L 2	105	0 R 21	0 L 10
2	0 L 3	1 L 2	106	0 R 61	1 R 42
3	0 L 5	1 L 2	107	0 R 121	1 R 8
4	1 R 6	0 L 4	108	0 R 101	1 R 122
5	0 L 9	1 L 2	109	0 L 207	0 L 95
6	0 R 6	0 R 10	110	0 R 124	1 R 104
7	0 R 189	1 R 159	111	0 R 115	1 L 72
8	0 R 14	1 R 13	112	0 R 194	1 R 180
9	0 L 15	1 L 2	113	0 R 160	1 R 183
10	0 L 16	0 R 10	114	0 R 148	1 R 146
11	0 L 26	1 R 128	115	0 R 11	1 R 129
12	0 R 19	1 R 20	116	0 L 131	0 L 1
13	0 L 89	1 L 31	117	0 R 39	1 R 72
14	0 L 11	0 R 22	118	0 R 61	1 R 132
15	0 L 23	1 L 2	119	0 R 61	1 R 133
16	1 L 1	0 L 1	120	0 R 70	0 L 16
17	0 R 24	0 R 32	121	0 R 14	1 R 132
18	0 R 98	1 R 113	122	0 R 21	1 R 134
19	0 R 28	1 R 29	123	0 R 151	1 R 188
20	0 R 30	1 R 31	124	0 R 137	1 L 20
21	0 R 24	0 R 59	125	0 R 130	1 R 179
22	0 R 33	1 R 22	126	0 R 203	1 R 156
23	0 L 34	1 L 2	127	0 R 75	1 R 175
24	0 L 35	0 R 0	128	0 R 75	1 R 205
25	0 R 36	1 R 25	129	0 R 202	1 R 157
26	0 R 37	1 R 38	130	0 R 12	1 R 141
27	0 R 39	0 L 40	131	0 R 30	1 R 69
28	0 R 41	1 R 42	132	0 R 39	1 L 124
29	0 R 43	1 R 44	133	0 R 153	0 L 16
30	0 R 45	1 R 46	134	0 R 154	0 L 16
31	0 R 21	1 R 47	135	0 R 88	1 R 136
32	0 R 48	1 R 32	136	0 R 196	1 R 140
33	0 R 32	1 R 33	137	0 R 24	0 R 48
34	0 L 74	1 L 2	138	0 R 99	1 R 37
35	1 R 35	1 L 1	139	0 R 171	1 R 148
36	0 R 50	1 R 36	140	0 R 96	1 R 159
37	0 R 17	1 R 111	141	0 R 174	1 R 165
38	0 R 13	1 R 52	142	0 R 12	1 R 159
39	0 L 11	0 R 36	143	0 R 88	1 L 165
40	0 L 4	1 L 53	144	0 R 88	1 R 158
41	0 R 24	0 R 25	145	0 L 152	1 L 163
42	0 R 13	0 L 16	146	0 R 84	1 R 88
43	0 R 55	1 R 56	147	0 R 159	1 R 12
44	0 R 57	1 R 58	148	0 R 83	1 R 143
45	0 R 24	0 R 54	149	0 R 144	1 R 155
46	0 R 60	1 R 61	150	0 L 66	1 L 141
47	0 R 62	0 L 16	151	0 R 150	1 R 169
48	0 R 25	1 R 48	152	0 R 159	1 R 114
49	0 L 35	0 R 63	153	0 R 97	1 R 97

Continued on the next page

q	Read 0	Read 1	q	Read 0	Read 1
50	0 R 16	1 R 50	154	0 R 49	0 R 32
51	0 R 147	1 R 143	155	0 R 191	1 R 187
52	0 R 67	1 R 68	156	0 L 194	1 R 194
53	0 L 40	1 L 53	157	0 L 186	1 R 96
54	0 R 22	1 R 54	158	0 R 167	1 R 178
55	0 L 11	0 R 50	159	0 R 19	1 R 170
56	0 R 60	0 L 16	160	0 R 164	1 R 186
57	0 R 69	1 R 70	161	0 R 159	1 R 96
58	0 R 61	1 R 71	162	0 R 125	1 R 164
59	0 R 54	1 R 59	163	0 R 97	1 R 162
60	0 R 49	0 R 25	164	0 L 189	1 R 142
61	0 L 28	1 L 72	165	0 R 146	1 R 171
62	0 L 117	1 L 30	166	0 R 96	1 R 123
63	0 R 73	1 R 63	167	0 R 161	1 R 143
64	0 R 97	1 R 150	168	0 R 148	1 R 87
65	1 R 184	1 L 53	169	0 R 161	1 R 96
66	0 R 144	1 R 145	170	0 R 180	1 R 122
67	0 R 76	1 R 208	171	0 L 192	1 R 60
68	0 L 196	1 R 31	172	0 R 145	1 R 88
69	0 R 41	1 R 62	173	0 R 152	1 R 201
70	0 L 89	1 R 78	174	0 R 140	1 R 84
71	0 R 55	1 L 79	175	0 R 174	1 R 75
72	0 R 80	1 R 81	176	0 R 126	1 R 168
73	1 R 0	1 R 63	177	0 R 138	1 R 116
74	0 R 74	1 R 82	178	0 R 97	1 R 51
75	0 L 145	1 L 141	179	0 R 167	1 R 51
76	0 R 85	1 R 86	180	0 R 17	1 L 131
77	0 R 20	1 R 38	181	0 R 143	1 R 149
78	0 R 89	1 L 132	182	0 R 76	1 R 116
79	0 L 108	1 R 56	183	0 R 106	1 R 124
80	0 R 49	0 R 33	184	1 L 40	1 L 109
81	0 R 91	0 L 42	185	0 R 174	1 R 156
82	0 R 177	1 R 77	186	0 R 181	1 R 182
83	0 R 96	1 R 12	187	0 R 66	1 R 204
84	0 R 88	1 R 97	188	0 R 140	1 R 197
85	0 R 206	1 R 21	189	0 R 84	1 R 190
86	0 R 116	1 R 100	190	0 R 157	1 R 185
87	0 R 173	1 R 166	191	0 R 64	1 R 7
88	0 R 102	1 R 103	192	0 R 150	1 R 127
89	0 R 80	1 R 79	193	0 R 18	1 R 94
90	0 R 101	1 R 195	194	0 L 200	1 R 172
91	0 R 105	1 R 106	195	0 R 104	1 R 96
92	0 R 65	0 R 0	196	0 R 90	1 R 88
93	0 R 27	1 R 107	197	0 R 139	1 R 199
94	0 R 182	0 L 207	198	0 R 150	1 R 112
95	1 L 109	1 L 95	199	0 R 201	1 R 200
96	0 R 19	1 R 110	200	0 R 156	1 R 198
97	0 R 60	1 R 60	201	0 L 123	1 R 167
98	0 R 68	1 R 135	202	0 R 196	1 R 28
99	0 R 93	1 R 108	203	0 R 163	1 R 192
100	0 R 105	0 L 207	204	0 R 146	1 R 176
101	0 R 117	1 R 118	205	0 R 161	1 R 193
102	0 R 71	1 R 58	206	0 R 45	1 R 100
103	0 R 57	1 R 119	207	0 L 4	0 L 207